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MODELLING LARGE OPEN ECONOMIES WITH INTERNATIONAL LINKAGES: THE USA AND EURO AREA

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SUMMARY

Empirical modelling of the linkages between the euro area and the USA requires an open economy framework. The methodology proposed in this paper achieves identification of a structural vector error correction model by supplementing restrictions from economic theory with assumptions for the direction of causality in cross-country contemporaneous relationships. Our baseline model assumes contemporaneous causality runs from the USA to the euro area for both output and inflation, with monetary policy domestically focused. The role of the USA as leading the euro area business cycle is reinforced by our results, but strong bidirectional cross-country interactions are uncovered for inflation and interest rates. Copyright © 2013 John Wiley & Sons, Ltd.

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1. INTRODUCTION

The USA and the euro area are the world's largest economies, together contributing a third of total world gross domestic product (GDP) and (excluding intra-euro area trade) accounting for more than a quarter of world exports.[1] Although events since 2008 have emphasized the crucial importance of linkages between these major blocks, the vast majority of empirical studies treat the USA and the euro area as closed economies. The present paper relaxes this unrealistic restriction by developing an empirically coherent vector autoregressive (VAR) structure which allows two-way feedback between them. Bayoumi and Swiston (2009, p. 356) make the point that there are a number of domestic economic relationships about which we are relatively comfortable, such as 'easily adjustable interest rates respond to shocks in more slow-moving inflation and output', but that causality for cross-economy linkages may be less easily determined a priori. Our solution is to employ variable-based ordering and make variable-specific cross-country contemporaneous causality assumptions, thereby clarifying the identification choices in this international context, while retaining the ability to maintain standard identification structures within each country.[2] In addition, we allow for the presence of long-run cross-country cointegrating relationships and distinguish permanent and temporary shocks.

The main contribution of this paper, therefore, is to model interactions between the USA and the euro area as two major open economies and hence compare their responses to domestic and international shocks. Our methodology is complementary to approaches recently used in an international context, such as the sign restriction VAR of Schneider and Fenz (2011), the factor structure of Canova and Ciccarelli (2009), the assumption of a single dominant unit in Pesaran and Chudik (2011) or the global vector autoregressive (GVAR) approach of Dees *et al*. (2007). While our approach has some cost in ignoring

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[1] These figures, which relate to 2010, are taken from the ECB website, at http://www.ecb.int/mopo/eaec/html/index.en.html.
[2] In what follows, we often loosely refer to the euro area as a 'country'.

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the rest of the world, modelling these two dominant economies together allows us to examine their interdependencies more explicitly than is feasible using, say, a factor or GVAR model. Nevertheless, although our context is two (major) economies only, our modelling has much in common with the international New Keynesian cointegrating GVAR model recently developed by Dees *et al*. (2011).

The paper proceeds as follows. Our methodology is presented in Section 2. Here we elucidate the usefulness of variable-based ordering and contemporaneous causality assumptions for identification in our US/euro area VAR, and also discuss issues arising from cointegration. This is followed by the details of our empirical specification in Section 3. Results are contained in Section 4, with these presented through an impulse response analysis in relation to domestic and foreign shocks. A final section concludes, with further details of the data and its characteristics provided in the Appendix.

2. METHODOLOGY

To facilitate discussion of identification for our two-country open-economy structural VAR, we adopt a variable-based ordering, where variables across different countries are listed by type. This contrasts with the usual country-based ordering employed in small open-economy VAR models (for example, Cushman and Zha, 1997), in which all variables for an individual country are grouped together, and with the sign restriction literature which has concentrated on identification using domestic responses to domestic shocks as in Schneider and Fenz (2011). When combined with assumptions about the economic structure, variable ordering facilitates consideration of how shocks are transmitted internationally because it does not require the same cross-country contemporaneous causal ordering to apply for different variables. This section discusses our methodology in the context of a structural system where variables may be either *I(0)* or *I(1)*.

2.1. Short-Run Identification Assumptions

To embody variable-based ordering, denote the US and euro area observations on a specific variable i (such as inflation) as $Y_{it} = \left(y_{it}^{\mathrm{US}}, y_{it}^{\mathrm{EA}}\right)'$. With bivariate vectors of observations on variables $i = 1, \ldots, n-1$ and adding the (scalar) bilateral exchange rate as variable $i = n$, the complete $m \times 1$ observation vector (where $m = 2(n-1)+1$) is

$$Y_t = \begin{bmatrix} Y_{1t} \\ Y_{2t} \\ \vdots \\ Y_{nt} \end{bmatrix} \tag{1}$$

It is important to note that the within-block order adopted here with the US first in each Y_{it} ($i = 1, \ldots, n\text{-}1$) is innocuous, as other assumptions are employed for identification purposes. Reflecting our focus on two large developed economies with broadly similar structures, we assume that the behavioural relationships have the same form (but not the same parameter values) for both the USA and the euro area.

Since some variables in the system are *I(1)*, we employ a vector error correction model (VECM) specification, which is (ignoring intercepts)

$$B(L)\Delta Y_t = \Pi Y_{t-1} + U_t \tag{2}$$

where Δ is the difference operator, $B(L) = B_0 - B_1 L - \ldots - B_p L^p$ (with L being the conventional lag operator) and B_0 is non-singular. The coefficient matrices B_ℓ ($\ell = 0, \ldots, p$) and Π are $m \times m$, while

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the $m \times 1$ zero-mean temporally uncorrelated disturbance vector U_t consists of n blocks, U_{it} $(i = 1, \ldots, n)$ partitioned as for Y_t in (1). Economic theory may provide identifying restrictions on the parameters of equation (2) for each economy, with these often being variable exclusion restrictions for B_ℓ (and particularly B_0), but does not necessarily provide within-block information that enables separation of the shocks applying to each economy. This situation can be characterized as equation (2) being a semi-structural VECM, such that the diagonal blocks of B_0 are 2×2 identity matrices and the covariance matrix of U_t has the form

$$E\left[U_{it}U_{jt'}\right] = \begin{cases} 0 & i \neq j \\ \Sigma_i & i = j \end{cases} \tag{3}$$

where 0 is a null matrix[3] of appropriate dimensions, Σ_i $(i = 1, \ldots, n\text{-}1)$ is a 2×2 positive definite covariance matrix while Σ_n is a positive scalar. Hence the blocks of U_t are mutually orthogonal, but the elements within a block may be correlated. Indeed, the multi-country New Keynesian model of Dees *et al*. (2011) embodies essentially the same covariance assumption as in equation (3).[4]

Dees *et al*. (2011) do not decompose the block covariances Σ_i. Here we employ assumptions about contemporaneous causality in order to obtain orthogonal shocks. This is distinct from block exogeneity, which is often imposed for small open economies (such as Canada in Cushman and Zha, 1997, or Australia in Dungey and Pagan, 2009). In contrast, our restrictions are imposed only on the contemporaneous relationships. Engle *et al*. (1983) refer to such restrictions as predetermined, but we shy away from using this term because of the timing conventions it infers.

Contemporaneous causality does not necessarily imply weak exogeneity, since there may be other endogenous variables entering the system such that simultaneity needs to be taken into account when estimating the parameters of these equations. Nevertheless, when the other identifying restrictions in equation (2) imply that the system is block recursive, then contemporaneous causality assumptions will result in B_0 having a recursive form. In that special case, weak exogeneity follows and ordinary least squares estimation can be applied equation by equation; see the discussion in Engle *et al*. (1983) relating to predetermined variables and weak exogeneity.

In a different vein, Pesaran and Chudik (2011, Assumption 2) apply an assumption which they refer to as dominance to decompose a cross-country covariance matrix such as Σ_i in a multi-country model. Their specification retains the influence of a specific country even as the number of countries in the model increases to infinity, whereas our focus is the modelling of the interrelationships across two large economies.

2.2. Permanent and Transitory Shocks

The structural vector error correction model (SVECM) of equation (2) allows the individual variables in Y_t to be either $I(0)$ or $I(1)$, since the $m \times m$ matrix $\Pi = \alpha\beta'$ incorporates the cointegrating relationships and error correction coefficients for $I(1)$ variables, together with aggregate effects for $I(0)$ variables.[5] Our modelling approach assumes that the US and euro area economies have similar structures, and hence both elements of any variable block Y_{it} in (1) are assumed to be of the same order of integration.

[3] Specifically, 0 is 2×2, except where i or $j = n$, when it is a zero row or column vector.

[4] The only difference between equation (3) and the corresponding assumption in Dees *et al*. (2011) is that they permit non-zero covariances between the exchange rate shocks and other blocks, with the real exchange rate modelled as an autoregressive process. However, our real exchange rate shocks are conditional on contemporaneous and lagged values of all other variables in the system (see Section 3 below).

[5] The column of β corresponding to an $I(0)$ variable contains all zeros except for a unit element relating to its own lag. Consequently, the dynamics of any $I(0)$ variable is written in terms of differences and its lagged level.

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The $I(1)$ variables in the system must be driven by one or more permanent shocks. With r cointegrating vectors (including those for stationary variables), each of α and β are $m \times r$ with rank r and it follows that the m-variable system for Y_t is subject to m-r permanent shocks and r transitory ones. As shown by Pagan and Pesaran (2008), the separation into permanent and transitory shocks can provide additional identifying information for an SVECM, when it is known that each of the permanent shocks is associated with a specific $I(1)$ variable. To consider this, define the $(m-r) \times m$ zero/one selection matrix S_1 such that $S_1 Y_t$ contains the elements of Y_t that are subject to permanent shocks. Further, define the $r \times m$ matrix S_2 that similarly selects the variables that are subject to transitory shocks only. Then S, of dimension $m \times m$, where

$$S = \begin{bmatrix} S_1 \\ S_2 \end{bmatrix}$$

reorders the elements of Y_t such that those subject to permanent shocks are ordered first in SY_t. By construction, S is non-singular.

Pre-multiplying the SVECM (2) by S, and also reordering the elements of Y_t, the system can be re-written as

$$G(L)\Delta(SY_t) = (S\alpha)\left(\beta' S^{-1}\right)(SY_{t-1}) + S\varepsilon_t \tag{4}$$

where $G(L) = G_0 - G_1 L - \ldots - G_p L^p$ with $G_i = SB_i S^{-1}$ $(i = 0, 1, \ldots, p)$ and ε_t is the relevant vector of orthogonal shocks. In the reordered system of equation (4), the error correction coefficient matrix can be partitioned as

$$S\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \tag{5}$$

where α_1 is the $(m-r) \times r$ submatrix corresponding to the first $m-r$ elements of SY_t, namely the elements of Y_t that experience permanent shocks. The moving average representation of equation (4) has the form

$$\Delta(SY_t) = J(L)(S\varepsilon_t) \tag{6}$$

where $J(L) = I_n + J_1 L + J_2 L^2 + \ldots$ is of infinite order and $J(1) = J$ has rank m-r. The permanent component (Y_t^p) of Y_t is then given by

$$\Delta\left(SY_t^p\right) = J(S\varepsilon_t) \tag{7}$$

where $(\beta' S^{-1})J = 0$. The final r columns of J in equation (7) must be zero since these multiply the transitory shocks which have no effect on Y_t^P, while the first m-r columns (multiplying the permanent shocks) must be non-zero and of rank m-r. Relating these conditions on J to the structural system (4), Pagan and Pesaran (2008) show that they require $\alpha_1 = 0$ in equation (5).

As discussed by Pagan and Pesaran (2008), information about the sources of permanent shocks restricts the appropriate specification of an SVECM, in particular implying that error correction coefficients should be set to zero in equations for variables that give rise to these permanent shocks. We follow this recommendation in our empirical analysis.

The considerations of Section 2.1 apply whether the corresponding shocks are both transitory or both permanent. Although the cross-country contemporaneous causality assumption could substantially affect

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the interpretation of the international sources of the permanent shocks when these arise from both elements of $Y_{it} = \left(y_{it}^{\mathrm{US}}, y_{it}^{\mathrm{EA}}\right)'$, nevertheless its importance is an empirical matter. A further relevant case may be that the elements of block i are $I(1)$ and cointegrated, but the corresponding (orthogonal) shocks consist of one permanent and one transitory shock. This situation will arise, for example, when output is cointegrated across economies and driven by a single permanent technology shock. It may then be implausible to assume that the permanent shock originates in one country (say the USA) but that the other country (the euro area) is causally prior for output. With the exception of this case, however, contemporaneous causality can be considered separately from the source(s) of permanent shocks.

3. EMPIRICAL SPECIFICATION

Having considered econometric issues, we now detail the specification employed for modelling the interactions between the USA and the euro area. For each economy, the framework consists of equations for output and inflation, both of which allow international influences, together with a monetary policy reaction function, with the model closed by a bilateral real exchange rate equation. This framework is similar to that used by Dees *et al*. (2011), but we apply contemporaneous causality assumptions with the aim of achieving a diagonal covariance matrix for U_t. In addition, other structural restrictions are imposed on current and lagged coefficients of our system; as discussed by Zha (1999), such restrictions can be important to generate economically sensible results.

The discussion below first considers the nature of the contemporaneous relationships and short-run dynamics (Section 3.1) and cointegration (Section 3.2), both in the context of a semi-structural VECM as in equations (2) and (3). Our contemporaneous causality assumptions are then discussed in Section 3.3.

3.1. Short-Run Dynamics

Adopting variable-based ordering as proposed in Section 2, the observation vector employed in our empirical model is written as $Y_t = \left(x_t^{\mathrm{US}}, x_t^{\mathrm{EA}}, \pi_t^{\mathrm{US}}, \pi_t^{\mathrm{EA}}, r_t^{\mathrm{US}}, r_t^{\mathrm{EA}}, q_t\right)'$ representing output, inflation and short-term interest rates (in that order) for each of the two economies, while q_t is the bilateral real exchange rate between the US dollar and the euro. Output variables and the real exchange rate are expressed in logarithms.

Our semi-structural VECM employs the conventional causal ordering assumptions across variables, and hence the contemporaneous coefficient matrix is lower block triangular. That is, this matrix has the form

$$B_0 = \begin{array}{c} \left[\begin{array}{ccccccccccc} 1 & b_{12}^0 & \vdots & 0 & 0 & \vdots & 0 & 0 & \vdots & 0 \\ b_{21}^0 & 1 & \vdots & 0 & 0 & \vdots & 0 & 0 & \vdots & 0 \\ \cdots & \cdots & & \cdots & \cdots & & \cdots & \cdots & & \cdots \\ b_{31}^0 & 0 & \vdots & 1 & b_{34}^0 & & \vdots & 0 & 0 & 0 \\ 0 & b_{42}^0 & \vdots & b_{43}^0 & 1 & \vdots & 0 & 0 & \vdots & 0 \\ \cdots & \cdots & & \cdots & \cdots & & \cdots & \cdots & & \cdots \\ b_{51}^0 & 0 & \vdots & b_{53}^0 & 0 & \vdots & 1 & 0 & \vdots & 0 \\ 0 & b_{62}^0 & \vdots & 0 & b_{64}^0 & \vdots & 0 & 1 & \vdots & 0 \\ \cdots & \cdots & & \cdots & \cdots & & \cdots & \cdots & & \cdots \\ b_{71}^0 & b_{72}^0 & \vdots & b_{73}^0 & b_{74}^0 & \vdots & b_{75}^0 & b_{76}^0 & \vdots & 1 \end{array}\right] \end{array} \begin{array}{l} \\ x_t \text{ block} \\ \\ \pi_t \text{ block} \\ \\ \\ r_t \text{ block} \\ \\ \\ q_t \text{ block} \end{array} . \tag{8}$$

Further, the lag coefficient matrices for each $\ell = 1, \ldots, p$ are of the form

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$$B_\ell = \begin{bmatrix} b_{11}^\ell & b_{12}^\ell & \vdots & -b_{15}^\ell & 0 & \vdots & b_{15}^\ell & 0 & \vdots & 0 \\ b_{21}^\ell & b_{22}^\ell & \vdots & 0 & -b_{26}^\ell & \vdots & 0 & b_{26}^\ell & \vdots & 0 \\ \cdots & \cdots & & \cdots & \cdots & & \cdots & \cdots & & \cdots \\ b_{31}^\ell & 0 & \vdots & b_{33}^\ell & b_{34}^\ell & \vdots & b_{35}^\ell & 0 & \vdots & b_{37}^\ell \\ 0 & b_{42}^\ell & \vdots & b_{43}^\ell & b_{44}^\ell & \vdots & 0 & b_{46}^\ell & \vdots & b_{47}^\ell \\ \cdots & \cdots & & \cdots & \cdots & & \cdots & \cdots & & \cdots \\ b_{51}^\ell & 0 & \vdots & b_{53}^\ell & 0 & \vdots & b_{55}^\ell & 0 & \vdots & 0 \\ 0 & b_{62}^\ell & \vdots & 0 & b_{64}^\ell & \vdots & 0 & b_{66}^\ell & \vdots & 0 \\ \cdots & \cdots & & \cdots & \cdots & & \cdots & \cdots & & \cdots \\ b_{71}^\ell & b_{72}^\ell & \vdots & b_{73}^\ell & b_{74}^\ell & \vdots & b_{75}^\ell & b_{76}^\ell & \vdots & b_{77}^\ell \end{bmatrix} \begin{matrix} x_t \text{ block} \\ \\ \\ \pi_t \text{ block} \\ \\ \\ r_t \text{ block} \\ \\ \\ q_t \text{ block} \end{matrix} \qquad \ell = 1, \ldots, p \tag{9}$$

where x_t, π_t and r_t are all 2×1 vectors. Thus our system imposes a number of restrictions in both the contemporaneous and lag coefficient matrices. As illustrated by Zha (1999), the failure to apply appropriate restrictions to lag matrices in a structural framework can lead to models with implausible economic implications. These dangers are particularly important in our case, since (in the seven-variable system) many coefficients would be estimated in an unrestricted VAR.

As implied by equation (8), output is assumed to be causally prior to all other variables and hence responds only to lagged values of these. However, unrestricted b_{12}^ℓ and b_{21}^ℓ in equation (9) allow short-run international business cycle linkages to run in both directions over time. On the other hand, the zero restrictions applied for the elements b_{14}^ℓ and b_{23}^ℓ together with the restrictions implied by setting $b_{13}^\ell = -b_{15}^\ell$ and $b_{24}^\ell = -b_{26}^\ell$ in equation (9) imply that the domestic (but not the foreign) real interest rate affects output. With the exchange rate also playing no role,[6] this open-economy structural specification for output has a similar form to that used by Milani (2010).

The inflation block does not permit foreign output to directly enter the international Phillips curve, in line with the empirical findings of Ihrig *et al*. (2007) and also Milani (2010). Note, however, that foreign output can affect domestic output, and domestic output in turn contemporaneously influences domestic inflation. Therefore, foreign output can play an indirect role. Lagged foreign inflation is included in equation (9) to capture international inflation spillovers, reflecting the findings of Dees *et al*. (2009) that foreign inflation contains important information in the context of a structural model of inflation in developed economies. The real exchange rate enters due to incomplete pass-through (Monacelli, (2005).

Monetary policy in equations (8) and (9) implies that interest rates are determined in relation to domestic economic conditions only. This is reflected in the block for $r_t = \left(r_t^{\mathrm{US}}, r_t^{\mathrm{EA}}\right)'$, experiencing no direct influence from current or lagged foreign output or inflation, nor the exchange rate.

Finally, the real exchange rate responds to developments in output, inflation and interest rates in both economies. In line with the view of Dees *et al*. (2011), we note that restrictions based on purchasing power parity or uncovered interest parity can apply if these are data-compatible, but they are not imposed in our specification for Δq_t.

3.2. Cointegration Issues

As discussed in Section 4 below, our system contains three *I(1)* variables, namely output in the two economies $\left(x_t^{\mathrm{US}}, x_t^{\mathrm{EA}}\right)$ and the real exchange rate q_t, which are linked by a single cointegrating vector, while the elements of π_t and r_t are assumed to be stationary. Thus our system with $m = 7$ variables

[6] Many models that allow incomplete pass-through (such as Monacelli, 2005) would allow a real exchange rate effect for output. However, this leads to economically implausible results in our case.

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is subject to two permanent shocks. These permanent shocks are assumed to originate in the output variables, x_t^{US} and x_t^{EA}, representing two distinct technology shocks. While theoretical international models often assume a common cross-economy technology shock, this does not seem to be empirically supported by the consistent deviations between growth rates experienced by different countries. Indeed, Uhlig (2010) finds differences in technological innovations to be the primary explanator for different monetary policy outcomes in the euro area and the USA and, in a cointegration analysis of output linkages for the USA, Europe and Japan, Papanyan (2010) finds two permanent shocks, one of which originates in the USA. Further, the assumption of distinct technology shocks is more palatable than considering exchange rate shocks as permanent; see also Dungey and Pagan (2009). Consequently, we assume that the real exchange rate shocks are transitory and this variable undertakes the adjustment required for the international cointegrating relationship to hold.

As discussed in Section 2, the error correction coefficients for variables experiencing permanent shocks should be set to zero (Pagan and Pesaran, 2008). Therefore, no error correction applies to the elements of $x_t = \left(x_t^{\mathrm{US}}, x_t^{\mathrm{EA}}\right)'$ in our empirical specification.

A further issue arises in relation to output 'gap' variables, which are defined as $x_t - x_t^P$, where x_t^P is the 2×1 non-stationary vector containing the permanent components of output, as given by equation (7). As widely recognized in macroeconomic modelling, these output gaps are appropriate for modelling inflation and interest rates. On the one hand, the use of essentially arbitrary data transformations, such as the application of the Hodrick–Prescott filter or the use of residuals from a deterministic trend, is unsatisfactory in ignoring the information provided by the cointegrating relationships. On the other hand, the estimation of the output gap within a fully specified SVECM substantially complicates model estimation. The solution of Dees *et al.* (2011) is to construct output gaps as deviations from long-run forecasts given by a reduced-form model that recognizes the unit root and cointegration properties of the data. However, we take the simpler solution proposed by Dungey and Pagan (2009), which approximates output gap effects by using current and lagged Δx_t together with the error correction term $\beta' Y_{t-1}$ that captures the cointegrating vector. This approach to approximating the output gap is employed in the inflation and interest rate equations.

These considerations lead to our specification of the error correction coefficient vector α of (5) as $\alpha = (0, 0, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7)'$. Note that no variable reordering of the form discussed in Section 2.2 is required since the permanent output shocks are ordered first within Y_t.

3.3. Contemporaneous Causality

The specification of our SVECM, given by the coefficient matrices (8) and (9), together with Π as defined by the cointegration relationship and specification of the error correction coefficients, is completed by contemporaneous causality assumptions. The assumption that monetary policy is domestically determined also implies that the disturbances applying to the US and euro area monetary policy reaction functions are uncorrelated, so that no further assumption is required for the interest rate block. Since permanent shocks apply to both output equations, with all other shocks being transitory, the potential complications with respect to the compatibility of permanent shocks and contemporaneous causality assumptions (noted at the end of Section 2.2) do not apply in this specification.

Therefore, contemporaneous causality assumptions are required for the output and inflation blocks of B_0 in equation (8), leading to four possible combinations. In the case of output there is empirical evidence that the USA is less affected by external factors than euro area countries (see, for example, Perez *et al.*, 2006; Kose *et al.*, 2008; Papanyan, 2010). Although Bayoumi and Swiston (2009) suggest that a strong stand should not be taken in relation to the source of

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output shocks in this international context, our experiments (not shown) imply that this assumption has relatively little qualitative effect on results.[7] Our structural specification imposes the restriction $b_{12}^0 = 0$ in equation (8), so that the permanent output shock corresponding to the euro area is obtained conditionally on that for the USA.

In contrast, despite substantial recent evidence that international inflation linkages are strong (see, for example, Wang and Wen, 2007; Ciccarelli and Mojon, 2010; Mumtaz and Surico, 2011; Bataa *et al*., 2012), the source(s) of these apparently common movements is unclear. Therefore, we entertain both contemporaneous causality assumptions for inflation. That is, the cases of $b_{34}^0 = 0$ and $b_{43}^0 = 0$ are both investigated, with the former, when the USA is contemporaneously causal, taken as the baseline case.

4. RESULTS

The first subsection briefly discusses our data (with further details in the Appendix) and cointegration results, before examining the properties of our model through a conventional impulse response analysis. While the main focus of this discussion is our baseline model with the USA contemporaneously causal over the euro area for both output and inflation, the case of contemporaneous causality for euro area inflation over the USA is also discussed.

4.1. Data and Cointegration

Our US/euro area model is estimated on quarterly data over 1983:Q1 to 2007:Q4, which represents the period of the move towards, and establishment of, a common currency in Europe and follows the Volker experiment period in the USA.[8] This sample period also avoids any unusual relationships that may have applied during the recent world recession. As usual with quarterly data, output is represented by (log) GDP, while the real exchange rate is expressed as US dollars per euro. Inflation is measured as the annual percentage change in the consumer price index (CPI), with three-month rates used for interest rates. Further details are provided in the Appendix, which includes graphs of the series, together with unit root and cointegration test results.

The classification of US and euro area log output, together with the real exchange rate, as $I(1)$ variables is well supported by unit root tests, while there is some evidence that inflation is stationary over this period; see Appendix Table A.I[9] Although tests point to interest rates being nonstationary, this may be a statistical consequence of interest rate smoothing by the central bank leading to sluggish responses of monetary policy (see Sack and Wieland, 2000). Our assumption of stationary inflation and interest rates is in line with the vast majority of macroeconomic analyses undertaken in the context of modern monetary policy.

Testing for cointegration is complicated by the univariate properties of the variables in our model. As discussed further in the Appendix, there is evidence for a single cointegrating vector amongst the $I(1)$ variables, and estimation by ordinary least squares yields

[7] This is due to the contemporaneous correlations between the output disturbances being relatively small in the semi-structural VECM of equations (2) and (3).

[8] The European Monetary System was established in March 1979, while the Volker experiment of targeting extended from 1979 to 1982.

[9] Halunga *et al*. (2009) illustrate that the conclusion on the order of integration for US and UK CPI inflation varies with the specific time period chosen and conclude that US inflation changed from $I(1)$ to $I(0)$ in 1982. This finding influenced the choice of 2003 as the starting date for our sample.

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$$x_t^{\mathrm{EA}} = 0.7323x_t^{\mathrm{US}} + 0.0362q_t + 7.3708 + \mathrm{ecm}_t \tag{10}$$

where ecm_t is the estimated disequilibrium at time t. Note that equation (10) implies differential trend rates of output growth in the two economies, a feature also found by Papanyan (2010) and indicated by the data graph in the Appendix.

4.2. Impulse Response Analysis

We begin with a brief summary of the main results. First, the model finds strong interest rate responses to inflationary impetus in each economy; the US response is slightly stronger, while euro area interest rates respond faster and for a more extended period.

Second, the cross-economy effects of shocks in output are consistent with the existing literature. US output shocks cause a significant response in euro area output, but the converse effect is insignificant. Foreign inflation shocks, however, have effects on inflation in both countries, and there are distinctly different monetary policy reactions in each country to foreign sourced inflation shocks from those generated domestically.

Finally, shocks to foreign interest rates (monetary policy) have significant effects on domestic interest rates. These effects are channelled via the cross-country inflation transmissions, and result from the foreign monetary policy stance and the already noted strong domestic monetary policy responses to changes in inflation.

All results are obtained from a VAR specification using $p = 3$ lags in the levels of all variables, corresponding to two lags in the difference VECM notation of equation (2). The model is estimated by ordinary least squares, conditional on the cointegrating relationship of equation (10) and the contemporaneous causality assumptions. Although the specification does not guarantee that the residuals are orthogonal, Table A.III in the Appendix verifies that the residuals are, indeed, effectively uncorrelated. Results are discussed first in relation to domestic and real exchange rate responses, before turning to the responses of both the USA and euro area to shocks originating in the other economy.

To facilitate cross-country comparisons, shocks to euro area variables are rescaled to be equal to the shock for the corresponding US variable, while shocks to US variables and the real exchange rate are equal to one residual standard deviation. The resulting shocks to output, inflation, interest rates and the real exchange rate are 0.0046, 0.49, 0.34 and 0.037 respectively. Error bands for the impulse responses are estimated using 5000 random draws (with replacement) from the estimated residuals, with the bands shown covering 68% of the bootstrapped responses. Statements relating to significance in the discussion below refer to these bands. Note that the bootstrap resamples only from stationary shocks; as shown by Berkowitz and Kilian (2000), such a procedure is valid when applied to the shocks driving $I(1)$ variables.

4.2.1. Domestic and Real Exchange Rate Responses

Panel I of Figure 1 shows key domestic responses. The significant permanent output responses to domestic output shocks are apparent in graphs (a) and (b). The long-run response to an own output shock is larger in the USA than in the euro area. These shocks also induce an inflationary response in each country (not shown) to which there is a strong monetary policy response, shown by the significant rises in interest rates in graphs (c) and (d). Initially the interest rate responses to domestic output shocks are similar for both countries, but the euro area response is sustained for a considerably longer period. The response does, however, return to zero in the longer run. The extended impulse responses to some temporary shocks are a feature of models using the Pagan and Pesaran (2008) identification strategy; see also Dungey and Fry (2009).

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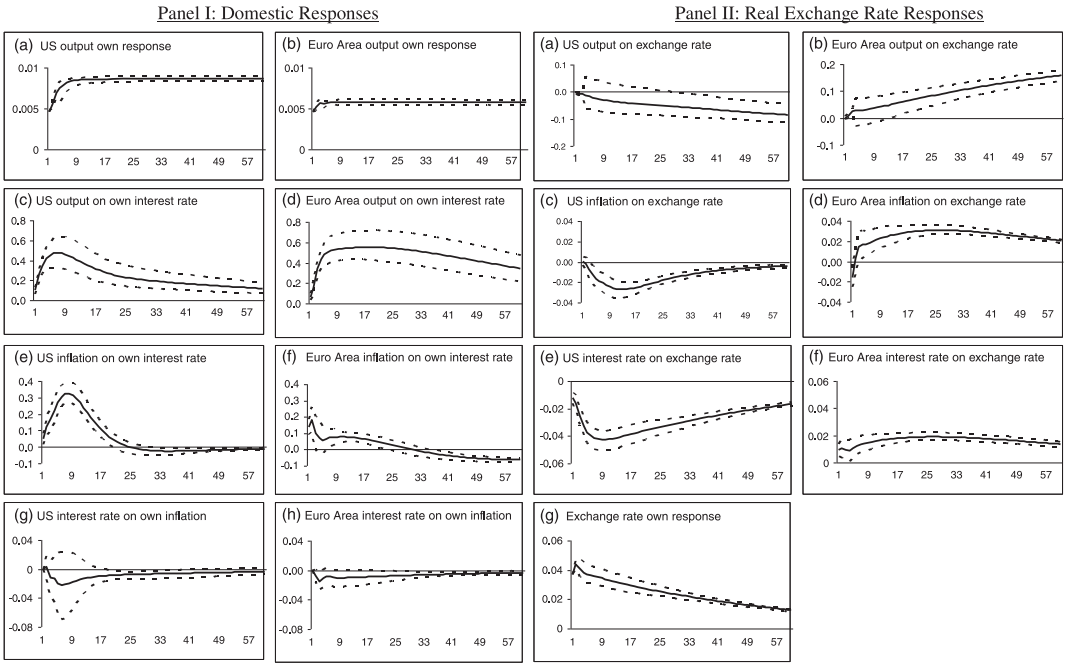

Figure 1. Domestic and exchange rate responses: impulse responses are shown as unbroken black lines with 68% confidence bands (obtained from 5000 bootstrap replications) shown as dashed lines. Euro area shocks are scaled to equal the corresponding US shock. Contemporaneous causality runs from the USA to the euro area for both output and inflation

Domestic inflation shocks in either country lead to increased interest rates as shown in (e) and (f) of panel I. Initially, the euro area responds more strongly to a domestic inflation shock than does the USA, but both effects are significant. However, the US response becomes stronger in the first several years and then rapidly reduces, whereas the euro area response tends to reduce slowly over a more protracted period.[10]

Domestic interest rate shocks, which we interpret as monetary policy shocks, lead to reductions in the inflation rates in their respective economies, graphs (g) and (h), although the error bands are wide for the USA at short horizons. By contrast, the GVAR model of Dees *et al*. (2007, Figure 3) results in a short-lived price puzzle for the USA.

Panel II of Figure 1 shows the effect of output, inflation and interest rate shocks in each economy on the real exchange rate. All responses are significant. In particular, a permanent output increase leads to a real appreciation for the currency concerned in (a) and (b). Significant appreciation also results from either inflation or interest rate shocks in graphs (c)–(f) of panel II. Although interest rates may be anticipated to lead to a currency appreciation, the effect of inflation on the real exchange rate is unclear a priori. Finally, (g) shows that exchange rate shocks are persistent; exchange rate shocks and responses tend to have extended, although transitory, effects in the system.

4.2.2. Domestic Responses to Foreign Shocks

The response of each economy to shocks in the other is of critical interest in this paper. Figure 2 shows relevant responses for our baseline model.

[10] The negative impulse response of euro area inflation to its own interest rate shock in the later period of the impulse response in Figure 1, panel I (f) is not permanent, but is a further example of the long-lived nature of the responses.

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With US output assumed to contemporaneously cause euro area output, a permanent US output shock leads to an increase in euro area output in graph (a) of panel I, while the US response to a corresponding euro area shock is insignificantly negative. These results are in line with those of Papanyan (2010). While Schneider and Fenz (2011) find an insignificant cross-economy effect of a euro area demand shock on the USA, in their case a US demand shock also has insignificant effects on euro area GDP. While our model implies that domestic inflation arising from foreign output shocks is not significant, graphs (c) and (d), it is sufficient to induce a significant change in interest rates in the USA, graphs (e) and (f). This is a reflection of the strong responses in US interest rates to any inflationary

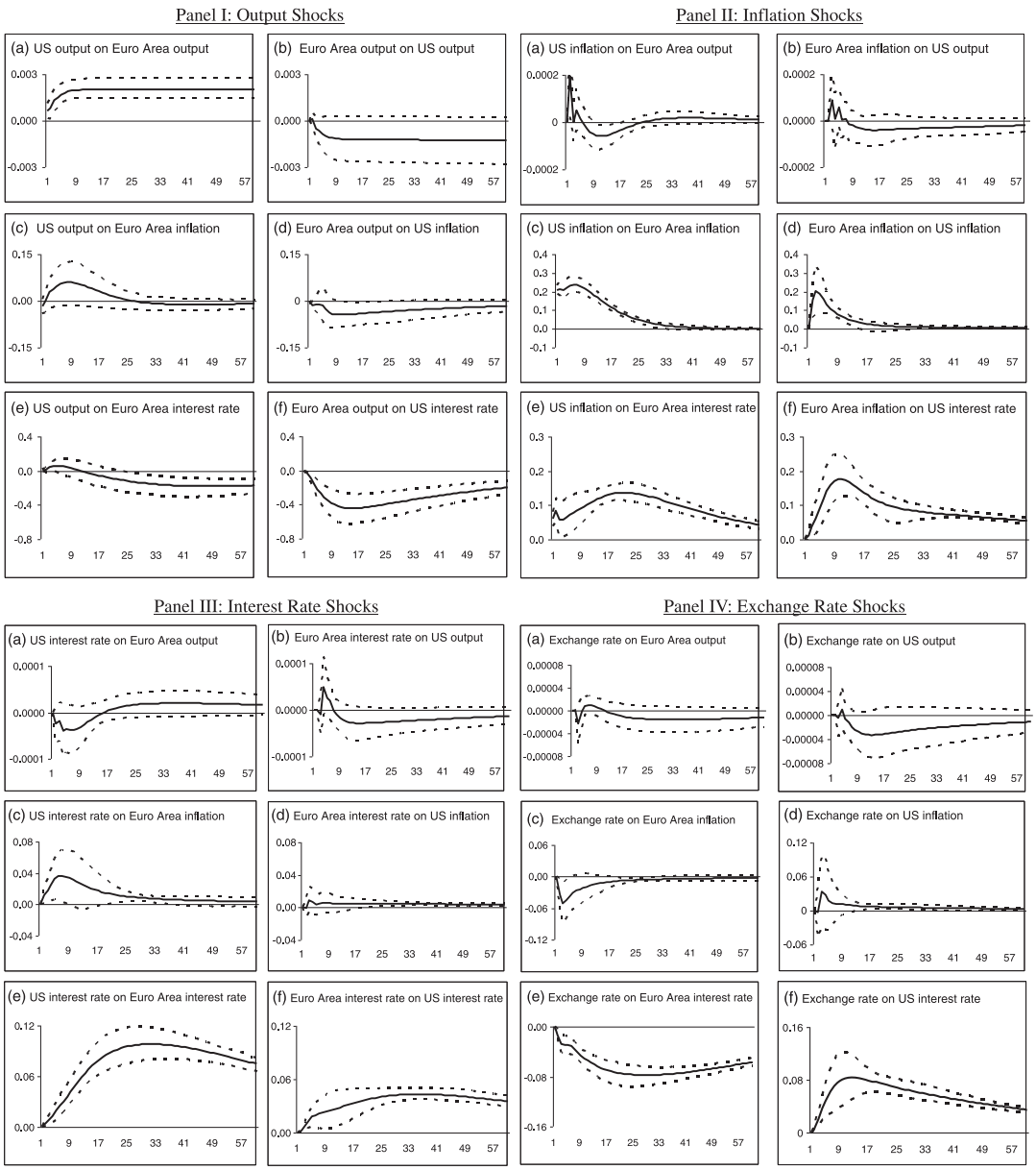

Figure 2. Domestic responses to foreign shocks (see Figure 1 for notes)

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signal in the US inflation rate—as discussed in the previous subsection—although in this case the mechanism for lower inflation is through an insignificant drop in US output.

Panel II of Figure 2 indicates that inflation has little effect on foreign output, but it plays a key role in domestic inflationary experiences. Inflation in the euro area rises sharply in response to a US inflation shock, partly reflecting that US inflation is assumed to contemporaneously cause that for the euro area. However, US inflation rises almost as much in response to a euro area inflation shock, albeit with some delay. Alternatively, this strong international transmission of inflation may be due to genuinely common international shocks; Ciccarelli and Mojon (2010) find that a common factor accounts for nearly 70% of inflation variance across 22 OECD countries. One candidate for this commonality is oil price inflation, which Dees *et al*. (2007) find significantly positively affects both US and euro area inflation. However, Kilian (2008) finds that the absence of oil supply disruptions would not have substantially altered the path of CPI inflation in G7 economies, and including oil price inflation in our model had no substantive effect on the results.

In response to higher inflation, each economy raises its interest rates. But in comparison with domestically sourced inflationary shocks, the interest rate responses to foreign sourced inflation are smaller and occur later than comparable interest rate responses to domestically sourced inflation. This reflects the importance of domestic conditions to monetary policy reactions; compare graphs (c) and (d) of Figure 2, panel III, with graphs (e) and (f) in panel I of Figure 1.

Panel III of Figure 2 shows cross-country responses to interest rate shocks. Interest rate increases in one country result in insignificant changes in output in the other country, which is similar to the GDP responses to monetary policy shocks in Schneider and Fenz (2011). However, higher US interest rates lead to an appreciation of the US dollar which, *ceteris paribus*, leads to an increase in inflation in the euro area through imported goods. The higher euro area inflation triggers a strong monetary policy response in the euro area, in the same manner as shown in the previous subsection. By a similar mechanism, higher euro area interest rates lead to an appreciation of the euro which, *ceteris paribus*, leads to increased US inflation and an increase in US interest rates. It may also be noted that although the significance appears to differ, a broadly similar pattern is found by Dees *et al*. (2007, Figure 3) for the response of euro area interest rates to a US shock.

Finally, Panel IV shows the effects of real exchange rate shocks on both economies. Once again, the output and inflation responses are relatively modest and insignificant (in relation to the confidence bands) at almost all lags, whereas interest rates respond in the anticipated way. That is, a real exchange rate shock (an appreciation of the euro) causes euro area interest rates to decline and US interest rates to increase.

4.2.3. Sensitivity Analysis

We conducted sensitivity analysis to the contemporaneous causality assumption on inflation by estimating the inflation equations of our model assuming euro area inflation is contemporaneously causal and repeating the impulse response analysis.[11] The shocks are normalized to have the same magnitude as those employed in the baseline specification.

As might be anticipated, when euro area inflation contemporaneously causes US inflation, euro area inflation shocks play a much greater role for the US economy than in the baseline case. In particular, the euro area inflation shock has a larger and quicker impact on US inflation, leading to a faster and bigger US monetary policy response. Conversely, of course, a US inflation shock has a slower and smaller impact on euro area inflation.

[11] In fact, we estimated SVECM models for all four cases, namely: (i) USA contemporaneously causes both euro area output and inflation; (ii) US contemporaneously causes euro area output and euro area contemporaneously causes US inflation; (iii) euro area contemporaneously causes US output and the USA contemporaneously causes euro area inflation; (iv) euro area contemporaneously causes both output and inflation for the USA. Detailed results for all cases are available from the authors on request.

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The contemporaneous causality assumption, however, does not drive the entire model. Significant cross-country responses to inflation and interest rate shocks are obtained under both inflation assumptions and the overall impulse responses are qualitatively similar. This is important, because it shows that our key research agenda, of finding a means of identifying a two open-economy structural VAR model, can be achieved by the simple means of contemporaneous causality assumptions without the stability of the entire framework becoming critically dependent on the implementation of those assumptions.

5. CONCLUSIONS

Modelling the interactions between the USA and euro area in an empirically coherent way in an open-economy framework provides significant challenges, but may be extraordinarily important for policy formation. This paper proposes a methodology which models these linkages in a structural VAR framework, without imposing a strict exogeneity ordering between the two economies. Rather, our model achieves identification by restricting the direction of contemporaneous relationships between key variables, with both dynamic relationships and long-run cointegrating relationships providing bidirectional cross-economy feedback. A major innovation of the paper is the use of variable ordering, whereby we group like data series together across the countries and are able to retain within-country structural restrictions. The cross-country contemporaneous causal ordering assumptions may vary across the different blocks of variables—so that it is feasible to have contemporaneous causality in inflation running in one direction and that for output in another.

The results emphasize the linkages between these large developed economies. Indeed, two-way cross-country relationships apply for inflation and interest rates, despite monetary policy being domestically focused, while US output also plays an important role for euro area output growth. These findings underline the importance of recognizing that, although the USA is the largest world economy, it is not immune from international influences.

A second theme of our empirical results is the broad similarity of patterns of responses of both economies to shocks. There are, of course, some distinctive features, with the USA responding faster and more strongly to a domestic inflationary shock, while euro area output growth has a negative effect (albeit insignificant) on US growth. However, particularly in relation to inflation, interest rate and real exchange rate shocks, their responses are generally similar.

In further work, we aim to extend the framework to incorporate a greater number of economies. In particular, there are cases where it is natural to assume that the USA and/or the euro area can be treated as block exogenous over smaller economies (such as Canada or the Accession countries of the European Union), and hence the number of contemporaneous causality assumptions required will not necessarily grow exponentially with the number of countries considered. It may also be feasible to extend the analysis to the three 'economy' case of the USA, euro area and the rest of the world. Within our framework, it is also possible to expand the number of variables considered, especially with the imposition of restrictions on the lagged responses, as proposed by Zha (1999) and employed in this paper. However, implementation of these ideas remains a matter for future research.

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APPENDIX: DATA ISSUES

Data

Euro area GDP data are from the November 2008 Area Wide Model (AWM) database (developed within the European Central Bank; see Fagan *et al*., 2005). However, the construction of monetary series in the AWM database using fixed historical (GDP) weights is not attractive, as it may overweight contributions of countries such as Italy and Portugal for periods when these countries did not follow disciplined monetary policy as (for example) in Germany. Therefore, we adopt the historical HICP inflation and interest rate data calculated by Anderson *et al*. (2011) using a sliding weight mechanism developed in order to represent the progress towards monetary union. However, from February 1994 euro area HICP inflation data are used, while the interest rate is represented by the three-month Euribor rate from the beginning of 1992. Inflation is measured at an annual rate throughout.

The approach and sliding weights of Anderson *et al*. (2011) are also used to construct a bilateral nominal exchange rate series (US dollars per euro) for the period prior to the introduction of the euro in January 1999. The real exchange rate is then obtained using this and price indices constructed from the US and Euro Area inflation rates. US data are drawn directly from the FRED database, made available by the Federal Reserve Bank of St Louis. Annual US inflation is measured in relation to the CPI for all urban consumers, while the interest rate is the three-month Treasury Bill rate.

The output and real exchange rate data are expressed as natural logarithms prior to analysis. The data series used in our model are shown in Figure A.1.

A.2 Unit Roots and Cointegration

Table A.I provides ADF and Phillips–Perron unit root test results for each output, inflation and interest rate series, together with the real exchange rate.[12] Despite a marginal rejection at 5% for the Phillips–Perron test applied to x_t^{US}, the $I(1)$ classifications are reasonably clear for both output series and for the real exchange rate. For inflation, the $I(1)$ null is rejected for the euro area by both tests and by the Phillips–Perron test for the USA, with the analysis of Halunga *et al*. (2009) providing further support for the $I(0)$ assumption in the US case. Although these tests do not reject the $I(1)$ null for interest rates, the setting of monetary policy in relation to inflation and the output gap points to stationarity.

The cointegration properties of two systems of interest are investigated in Table A.II using the Johansen trace test. The first system consists of the three $I(1)$ variables, namely the two output series and the real exchange rate (all in logs), and the second system contains all seven variables included in the SVECM model of the text. The analysis allows unrestricted intercepts, thus permitting trends in the levels of the variables, which is clearly important for output; however, there is no clear economic rationale for including a separate trend variable in the VAR and hence this is not considered. The VAR models in Table A.II employ three lags in differences, to ensure the residuals are relatively free of

[12] This preliminary unit root and cointegration analysis has been conducted in EViews.

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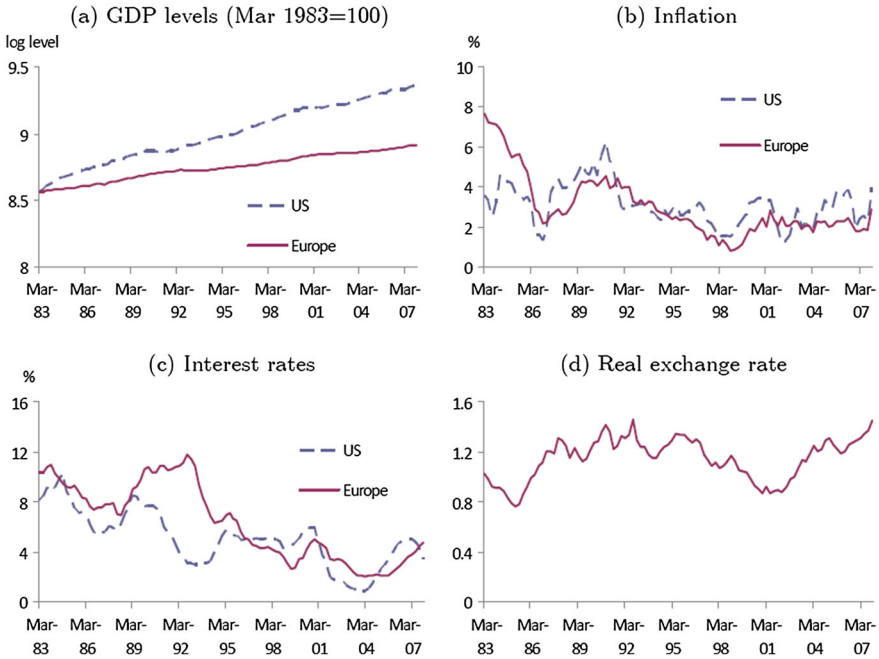

Figure A.1. Data series for the euro area and the USA

Table A.I Unit root test results

Series	ADF		Phillips–Perron	
x_t^{US}	−2.85	(0.185)	−3.50*	(0.045)
x_t^{EA}	−2.43	−1.87	(0.661)	
π_t^{US}	−2.18	(0.216)	2.98*	(0.041)
π_t^{EA}	−3.20*	(0.023)	−3.26*	(0.019)
r_t^{US}	−1.47	(0.546)	−1.89	(0.334)
r_t^{EA}	−1.58	(0.490)	−1.39	(0.582)
q_t	−2.89	(0.179)	−1.84 7	(0.361)

Note: ADF is the conventional augmented Dickey–Fuller test; augmentation is selected by AIC to a maximum of eight lags. PP is the Phillips–Perron test with Bartlett kernel and automatic Newey–West bandwidth selection. Values in parentheses are *p*-values. Tests for x_t^{US} and x_t^{EA} include intercept and trend; all others include intercept only. Asterisk indicates significance at *5%.

autocorrelation.[13] Although details are not shown, it is worth noting that cointegration between the two output series alone is not supported (with a *p*-value of 0.111). However, with the inclusion of the real exchange rate, the three-variable system provides evidence for a single cointegrating vector.

A simple reading of the results for the seven-variable system in Table A.II indicates the presence of three cointegrating relationships. However, these results are complicated by the statistical properties of the inflation and interest rate variables discussed above—specifically that they have characteristics similar to *I*(*1*) variables over our sample period. Although the *p*-values for the number of cointegrating

[13] It may be noted that two lags in differences are used in the empirical models of the text. The use of three lags there leads to some evidence of seasonal patterns in the impulse responses (despite the use of seasonally adjusted data), but are otherwise qualitatively similar to those reported.

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Table A.II. Johansen cointegration trace test results

	Three-variable system			Seven-variable system		
H_0	Ordered eigenvalues	Trace statistic	p-value	Ordered eigenvalues	Trace statistic	p-value
$r=0$	0.207	32.91	0.0212	0.564	207.66	0.0000
$r\leq 1$	0.093	10.67	0.2325	0.416	127.95	0.0001
$r\leq 2$	0.014	1.33	0.2496	0.303	76.24	0.0140
$r\leq 3$				0.182	41.54	0.1720
$r\leq 4$				0.158	22.31	0.2815
$r\leq 5$				0.049	5.85	0.7133
$r\leq 6$				0.011	1.02	0.3120

Note: The three-variable system consists of the two output series and the real exchange rate (all in logs), while the seven-variable system employs the set of variables discussed in the text. Both VARs include augmentation with three lags in differences and intercepts are unrestricted; p-values are obtained using the method of MacKinnon *et al*. (1999).

Table A.III. Residual correlations of baseline model

	x_t^{US}	x_t^{EA}	π_t^{US}	π_t^{EA}	r_t^{US}	r_t^{EA}	q_t
x_t^{US}	1.000						
x_t^{EA}	−0.024	1.000					
π_t^{US}	−0.023	0.090	1.000				
π_t^{EA}	0.058	0.004	0.018	1.000			
r_t^{US}	0.027	−0.060	0.022	0.009	1.000		
r_t^{EA}	0.072	−0.000	0.011	−0.032	0.063	1.000	
q_t	−0.035	−0.030	−0.028	0.025	0.031	0.027	1.000

Note: The baseline model assumes contemporaneous causality from the USA for both output and inflation.

vectors in this system point to 3, the magnitude of the eigenvalues drops relatively slowly for the first five, with the sixth eigenvalue being less than one third of the preceding value. In the context of a Johansen analysis, $r=5$ cointegrating relationships are anticipated for a system consisting of four stationary variables and a single cointegrating vector across three $I(1)$ variables; we are comfortable that Table II is compatible with such a system. When the five cointegrating vectors are identified by writing each of x_t^{EA}, π_t^{US}, π_t^{EA}, r_t^{US} and r_t^{EA} in terms of x_t^{US} and q, that for x_t^{EA} is qualitatively very similar to the Engle–Granger estimated relationship of equation (10). Further, the unrestricted adjustment coefficients for x_t^{US} and x_t^{EA} are not significant, whereas that for q is significant (t-ratios of 1.50, 0.67 and 2.34, respectively).

Residual Correlations

Table A.III shows the residual cross-correlations for the baseline model with both US output and inflation contemporaneously causal for the euro area. With the largest of these having a magnitude of 0.09, they are considered sufficiently small for the residuals to be treated as orthogonal. Very similar cross-correlations result when euro area inflation is assumed to be contemporaneously causal for US inflation (with US output remaining contemporaneously causal for euro area output).

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